

3. Equilibrium of a Particle

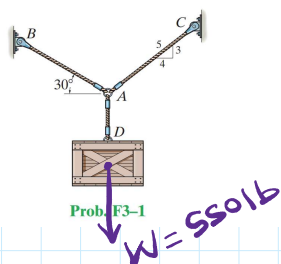
Tuesday, August 20, 2024 8:42 AM

* Analyze particles that are in equilibrium ($\vec{a} = 0$). Constant velocity or @ rest.

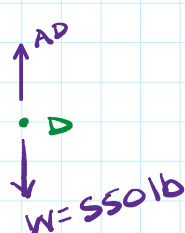
* $\vec{F} = m\vec{a} \rightarrow \Sigma \vec{F} = 0$

* Spring $\rightarrow F_s = k(\underline{l} - \underline{l}_0)$, FBD

F3-1. The crate has a weight of 550 lb. Determine the force in each supporting cable.



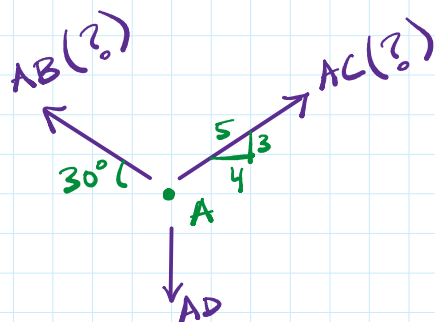
FBD



$\Sigma F = m\vec{a}$
 $\Sigma F_y = 0$

$AD - W = 0$
 $AD = W$

$AD = 550 \text{ lb}$



$\Sigma F_y = 0$ $\Sigma F_x = 0$

$AB_x = -AB \cos(30^\circ)$

$AB_y = AB \sin(30^\circ)$

$AC_x = \frac{4}{5} AC$

$AC_y = \frac{3}{5} AC$

$\Sigma F_x = 0$

$AC_x + AB_x = 0 \rightarrow \left(\frac{4}{5} AC - \frac{\sqrt{3}}{2} AB = 0 \right) \quad (1)$

$\Sigma F_y = 0$

$AC_y + AB_y + AD = 0 \rightarrow \left(\frac{3}{5} AC + \frac{1}{2} AB - 550 = 0 \right) \quad (2)$

$\frac{3}{5} AC + \frac{1}{2} AB = 550 \quad (2a)$

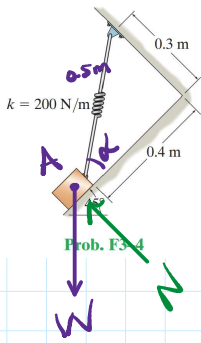
$$\begin{bmatrix} \frac{4}{5} & -\frac{\sqrt{3}}{2} \\ \frac{3}{5} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} AC \\ AB \end{bmatrix} = \begin{bmatrix} 0 \\ 550 \end{bmatrix}$$

$A \cdot X = b$

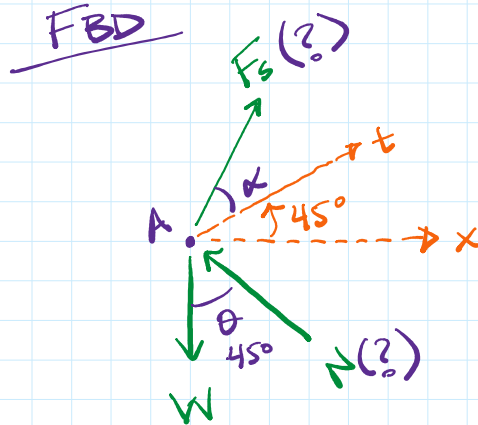
$AB = 478.46 \text{ lb}$

$AC = 516.74 \text{ lb}$

F3-4. The block has a mass of 5 kg and rests on the smooth plane. Determine the unstretched length of the spring.



$$W = mg = (5 \text{ kg}) (9.81 \text{ m/s}^2) = 49.0 \text{ N}$$



$$\alpha = \tan^{-1}\left(\frac{0.3}{0.4}\right)$$

$$\alpha = 36.86^\circ$$

$$\sum F_y = 0$$

$$F_s \sin(\alpha + 45^\circ) + N \cos(45^\circ) - W = 0$$

$$F_s \sin(81.86^\circ) + N \cos(45^\circ) = 49$$

$$\textcircled{1} - (0.99 F_s + 0.707 N = 49)$$

$$\sum F_x = 0$$

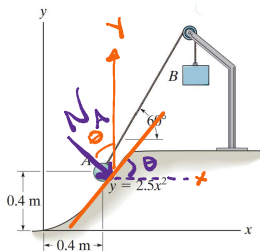
$$F_s \cos(\alpha + 45^\circ) - N \sin(45^\circ) = 0$$

$$\textcircled{2} - (0.14 F_s - 0.707 N = 0)$$

$$F_s = k(l - l_0)$$

↑ ↑ ↑ ?
200 0.5

3-37. A 4-kg sphere rests on the smooth parabolic surface. Determine the normal force it exerts on the surface and the mass m_B of block B needed to hold it in the equilibrium position shown.



Prob. 3-37

$$m = 4 \text{ kg} \quad W = mg$$

$$m_B = ? \quad N_A = ?$$

$$\tan \theta = \frac{2}{1}$$

$$\theta = \tan^{-1}(2)$$

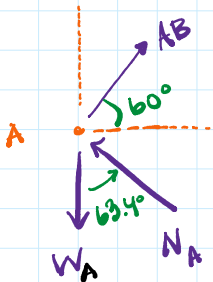
$$\frac{dy}{dx} = \frac{d}{dx}(2.5x^2) = 5x$$

$$\theta = 63.4^\circ$$

$$\frac{dy}{dx} = \frac{d}{dx}(2.5x^2) = 5x$$

$$\theta = 63.4^\circ$$

slope @ $x=0.4m$ $f'(0.4) = 5(0.4) = \frac{2}{1}$ - rise
1 - run



$$\sum F_y = 0$$

$$AB \sin(60^\circ) - W_A + N_A \cos(63.4^\circ) = 0$$

$$0.866(AB) - W_A + 0.45 N_A = 0$$

$$(0.866 \cdot AB + 0.45 N_A = W_A) \quad (1)$$

$$\sum F_x = 0$$

$$AB \cdot \cos(60^\circ) - N_A \cdot \sin(63.4^\circ) = 0$$

$$0.5(AB) - 0.89 N_A = 0 \quad (2)$$

$$W_A = (4 \text{ kg})(9.81 \text{ m/s}^2)$$

$$W_A = 39.24 \text{ N}$$

$$AB = 35 \text{ N}$$

$$N_A = 19.7 \text{ N}$$

$$AB = \frac{0.89 N_A}{0.5}$$

$$AB = 1.78 N_A \quad (2a)$$



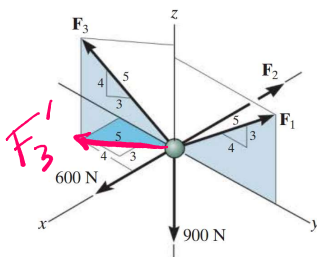
$$\sum F_y = 0$$

$$AB - W_B = 0 \rightarrow W_B = AB$$

$$m_B g = AB \rightarrow m_B = \frac{AB}{g} = \frac{35 \text{ N}}{9.81 \text{ m/s}^2}$$

$$m_B = 3.6 \text{ kg}$$

F3-7. Determine the magnitude of forces F_1 , F_2 , F_3 , so that the particle is held in equilibrium.



Prob. F3-7

Known Forces

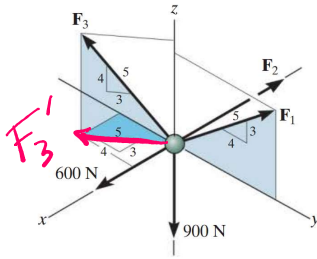
$$600 \hat{i} + 0 \hat{j} + 0 \hat{k} \text{ N}$$

$$0 \hat{i} + 0 \hat{j} - 900 \hat{k} \text{ N}$$

$$\vec{F}_1 = 0 \hat{i} + \frac{4}{5} F_1 \hat{j} + \frac{3}{5} F_1 \hat{k}$$

$$\vec{F}_2 = -F_2 \hat{i} + 0 \hat{j} + 0 \hat{k}$$

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Prob. F3-7

Known Forces

$$600 \hat{i} + 0 \hat{j} + 0 \hat{k} \text{ N}$$

$$0 \hat{i} + 0 \hat{j} - 900 \hat{k} \text{ N}$$

$$\vec{F}_1 = 0 \hat{i} + \frac{4}{5} F_1 \hat{j} + \frac{3}{5} F_1 \hat{k}$$

$$\vec{F}_2 = -F_2 \hat{i} + 0 \hat{j} + 0 \hat{k}$$

$$\vec{F}_3 = \frac{3}{5} \left(\frac{3}{5} F_3 \right) \hat{i} - \frac{4}{5} \left(\frac{3}{5} F_3 \right) \hat{j} + \frac{4}{5} F_3 \hat{k}$$

$$F_3' = \frac{3}{5} F_3$$

$$\sum F_x = 0$$

$$600 - F_2 + \frac{9}{25} F_3 = 0$$

$$\sum F_y = 0$$

$$\frac{4}{5} F_1 - \frac{12}{25} F_3 = 0$$

$$\sum F_z = 0$$

$$-900 + \frac{3}{5} F_1 + \frac{4}{5} F_3 = 0$$

$$F_1 = \frac{5}{4} \left(\frac{12}{25} \right) F_3 = \frac{3}{5} F_3$$

$$900 = \frac{3}{5} \left(\frac{3}{5} F_3 \right) + \frac{4}{5} F_3 \rightarrow 900 = \left(\frac{9}{25} + \frac{4}{5} \right) F_3$$

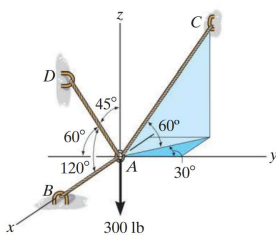
$$900 = \frac{29}{25} F_3 \rightarrow F_3 = 900 \left(\frac{25}{29} \right) \rightarrow F_3 = 775.86 \text{ N}$$

$$F_1 = \frac{3}{5} (775.86) \rightarrow F_1 = 465.52 \text{ N}$$

$$F_2 = 600 + \frac{9}{25} (775.86)$$

$$F_2 = 879.30 \text{ N}$$

F3-10. Determine the tension developed in cables AB, AC, and AD.



Prob. F3-10

$$AD_x = AD \cos(120^\circ) = -0.5 AD$$

$$AD_y = -AD \cos(60^\circ) = -0.5 AD$$

$$AD_z = AD \cos(45^\circ) = 0.707 AD$$

$$AC_x = -AC \cos(60^\circ) \sin(30^\circ) = -0.25 AC$$

$$AC_y = AC \cos(60^\circ) \cos(30^\circ) = 0.433 AC$$

$$AC_z = AC \sin(60^\circ) = 0.866 AC$$

$$AB_x = AB$$

$$AB_y = 0$$

$$AB_z = 0$$

$$AB_y = 0$$

$$AB_z = 0$$

$$\sum F_x = 0$$

$$(1) -0.5AD - 0.25AC + AB = 0$$

$$\sum F_y = 0$$

$$(2) -0.5AD + 0.433AC + 0 = 0 \rightarrow$$

$$AD = 0.866AC$$

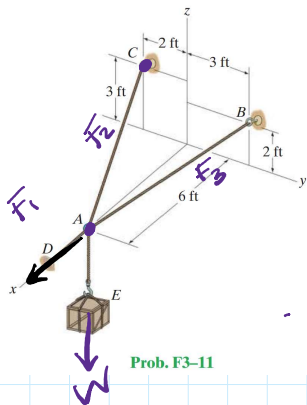
Substitute

$$\sum F_z = 0$$

$$(3) 0.707AD + 0.866AC + 0 = 300$$

$$AB = 139 \text{ lb} \quad AC = 203 \text{ lb} \quad AD = 176 \text{ lb}$$

F3-11. The 150-lb crate is supported by cables AB, AC, and AD. Determine the tension in these wires.



Prob. F3-11

Unit vectors

$$\underline{AC} = \frac{-6\hat{i} - 2\hat{j} + 3\hat{k}}{\sqrt{(-6)^2 + (-2)^2 + 3^2}}$$

$$AC = -\frac{6}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{3}{7}\hat{k}$$

$$\underline{F_2} = F_2 \cdot \left\langle -\frac{6}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{3}{7}\hat{k} \right\rangle$$

Unit Vector

$$\underline{AB} = \frac{-6\hat{i} + 3\hat{j} + 2\hat{k}}{\sqrt{(-6)^2 + (3)^2 + (2)^2}}$$

$$AB = -\frac{6}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{2}{7}\hat{k}$$

$$\underline{F_3} = F_3 \cdot \left\langle -\frac{6}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{2}{7}\hat{k} \right\rangle$$

$$\underline{\sum F_x = 0}$$

$$-6F - 6F - 6F - 6F$$

$$\underline{\sum F_y = 0}$$

$$-2F + 3F - 6F - 6F$$

$$\underline{\Sigma F_x = 0}$$

$$F_1 - \frac{6}{7} F_2 - \frac{6}{7} F_3 = 0$$

$$\underline{\Sigma F_y = 0}$$

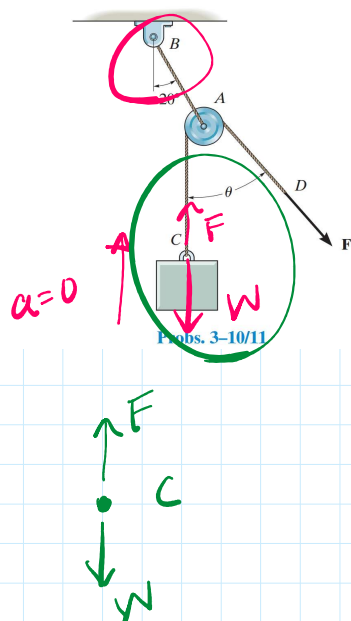
$$-\frac{2}{7} F_2 + \frac{3}{7} F_3 = 0$$

$$\underline{\Sigma F_z = 0}$$

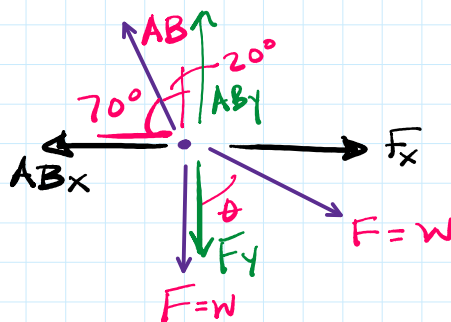
$$\frac{3}{7} F_2 + \frac{2}{7} F_3 - W = 0$$

3-10. The block has a weight of 20 lb and is being hoisted at uniform velocity. Determine the angle θ for equilibrium and the force in cord AB.

3-11. Determine the maximum weight W of the block that can be suspended in the position shown if cords AB and CAD can each support a maximum tension of 80 lb. Also, what is the angle θ for equilibrium?



FBD



$$\Sigma F_x = 0$$



$$\underline{AB \cos(70^\circ) = F \sin \theta}$$

$$\underline{AB \cos(70^\circ) = W \sin \theta} \quad (1)$$

$$\Sigma F_y = 0$$

$$AB \sin(70^\circ) = F + F \cos \theta$$

$$(2) \quad \underline{AB \sin(70^\circ) = W(1 + \cos \theta)}$$

$$AB = \frac{W \sin \theta}{\cos(70^\circ)}$$

W

$$AB \sin(70^\circ) = 1 + 1 \cos \theta$$

$$(2) \quad AB \sin(70^\circ) = W(1 + \cos \theta)$$

$$\cancel{W} \sin \theta \cdot \tan(70^\circ) = \cancel{W} (1 + \cos \theta)$$

$$\sin \theta \cdot \tan(70^\circ) = 1 + \cos \theta$$

Solved graphically

$$\theta = 0.7 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right)$$

$$\theta = 40.1^\circ$$

$$AB = \frac{20 \cdot \sin(40.1^\circ)}{\cos(70^\circ)}$$

$$AB = 37.67 \text{ lb}$$